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OSSP:

Commands:

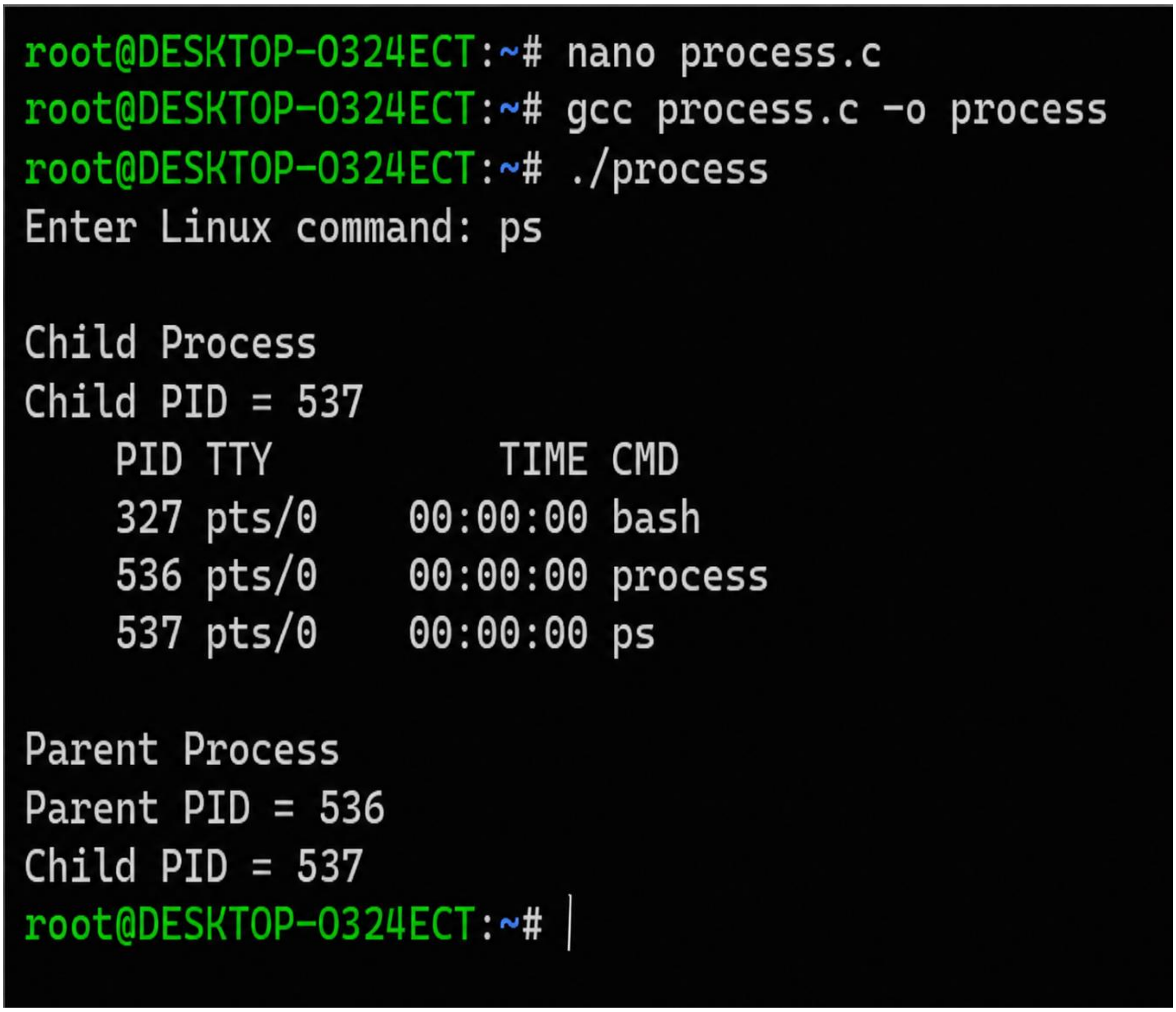
Uname : It displays the System information

Lscpu : It displays the complete CPU information

```
root@DESKTOP-0324ECT:~# nano process.c
root@DESKTOP-0324ECT:~# gcc process.c -o process
root@DESKTOP-0324ECT:~# ./process
Enter Linux command: lscpu

Child Process
Child PID = 523
Architecture:              x86_64
  CPU op-mode(s):          32-bit, 64-bit
  Address sizes:           39 bits physical, 48 bits virtual
  Byte Order:              Little Endian
CPU(s):                    8
  On-line CPU(s) list:     0-7
Vendor ID:                 GenuineIntel
  Model name:              11th Gen Intel(R) Core(TM) i7-1185G
                           7 @ 3.00GHz
    CPU family:            6
    Model:                 140
    Thread(s) per core:    2
    Core(s) per socket:    4
    Socket(s):             1
    Stepping:              1
    BogoMIPS:              5990.39
```

Ps : It displays the running process of our file



```
root@DESKTOP-0324ECT:~# nano process.c
root@DESKTOP-0324ECT:~# gcc process.c -o process
root@DESKTOP-0324ECT:~# ./process
Enter Linux command: ps

Child Process
Child PID = 537
    PID TTY          TIME CMD
    327 pts/0    00:00:00 bash
    536 pts/0    00:00:00 process
    537 pts/0    00:00:00 ps

Parent Process
Parent PID = 536
Child PID = 537
root@DESKTOP-0324ECT:~#
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among multiple programs by giving each one a turn to execute.
Example**:** When we open Chrome and a music player together, the OS schedules CPU time for both.It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

The operating system gives memory to each program that is running
It makes sure one program does not use another program's memory.
Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system storesand organizes files on the storage device. Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.